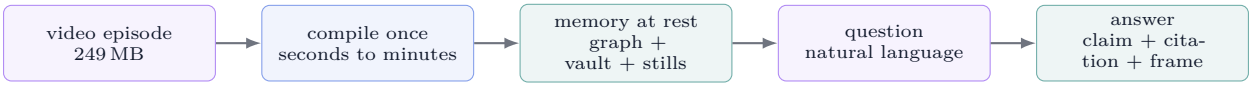
Chronicle
The short version: video-free memory

What the new architecture does, in five minutes — October 2026

One sentence. Chronicle compiles a video episode *once* into a compact timestamped memory — VLM captions, a salience event ledger, a few evidence stills and an LLM-ready pack — then answers questions from that memory in microseconds, with **no video and no video model in the query path**.

249 MB	95 KB	0.05–0.06 ms	0
source video	compiled memory	vault query, p50	video at answer time

The problem. Every question about video normally means decoding, sampling frames and calling a vision model again. That is slow, expensive, and coverage depends on which frames you happened to sample. Long episodes make it worse: the video is huge, but the part you need is tiny.



Three layers, three jobs.

Layer	What it is	What it answers
Graph	The index: entities, states, edges, zones, counts	where / state / who / how many
Vault	The memory: captions, coalesced events, metadata, cache	what happened / around when / describe
Evidence	480 px WebP stills + citations	show me / prove it

Key idea. The graph is the index; the captions and events are the recallable memory; the vault is the fast answer engine; evidence makes it auditable. Compile once, query many times.

1 What happens: compile once

This is the only expensive step. It runs once per clip, in the background, on first view. After it finishes, the source video is only needed for playback and screenshots.

1. **Snapshot the source DB.** Make a consistent copy of the graph while ingest may still be writing.
2. **Caption the video.** Cut it into 30 s windows (20 s for short clips); sample three frames per window; caption each frame with a VLM and synthesise a window caption. Store objects, actions, zone and tags.
3. **Build the event ledger.** Read state transitions, anchors and edges; remove visibility noise (pan, occlusion, reacquisition) and merge repeated touches/placements into memory-worthy events. Pack at about 17 bytes per event.
4. **Build the vault.** Write the packed events, string dictionary, captions, per-label count summaries, metadata and an answer cache into a small SQLite file.
5. **Extract evidence.** Save a bounded set of 480 px WebP stills: interaction strips, object crops and zone keyframes. No full-resolution frames are kept.
6. **Write the pack and report.** Produce an LLM-ready pack (2k/4k/8k token budgets) and a single-file HTML report with a timeline and query box.

What gets stored, and why it is small.

Artefact	What it contains	Typical
Graph DB	entities, states, edges, zones, observations (source of truth)	168–254 KB
Vault	packed events, captions, strings, count summaries, answer cache	28–32 KB
Caption ledger	30 s windows: caption, objects, actions, tags, 3 frame captions	1–2 windows
Event ledger	salience-coalesced state / edge / interaction events	8–11 events
Evidence	480 px WebP stills, ranked by interaction / crop / zone keyframe	8 frames
Memory pack	verified answer + ranked captions/events + evidence refs + rules	2.4–3.0 KB
Whole job dir	vault + ledger + evidence + report + pack + zip	0.99–1.3 MB

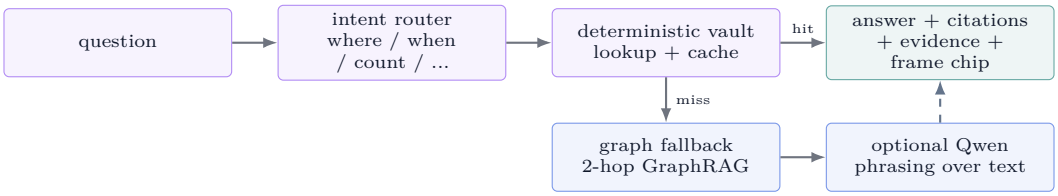
A 60-minute synthetic episode with 2,474 events, 184 caption windows and 40 snapshots produces a 142 KB vault. The source video is orders of magnitude larger.

Key idea. The memory is deliberately lossy. It does not preserve every pixel; it preserves enough *timestamped meaning* to answer repeated questions, plus a pointer back to the evidence.

Why this is more useful than the graph alone. The graph knows `person VISIBLE_MOVING`; the caption knows "a person walks on a sidewalk holding a cup of coffee". Captions also cover segments where the tracker emitted no interaction row, and their timestamps make "around 10 s" a range query rather than a similarity search over one frame.

2 How it is searched

Query time touches only small text structures. There is no video decoding, no frame sampling and no vision model in the default path.



What the router understands: where / when / around / before / after / count / holding / inventory / yes-no / keyword. It resolves the entity and time, looks up captions and events, and returns a deterministic answer with a citation. Paraphrases hit a canonical answer cache.

Trace: "What happened around 10 seconds?"

1. Parse the intent `around` and the time $t = 10$ s.
2. Find the caption windows and coalesced events covering that range in the vault.
3. Rank the nearest window/events and attach citation timestamps.
4. Compose the answer deterministically; cache the canonical form.
5. Attach an evidence still if one is within 8 s, and render the `frame @ t` chip.
6. If the vault has no match, ask the graph (state/edges) and label the answer `graph-fallback`.

Cost of each path (30-question internal regression suite):

Path	Accuracy	p50	Video at query
Deterministic vault	**30/30**	**0.05–0.06 ms**	no
Qwen 7B over the memory pack	**30/30**	1.07 s	no
Original graph + Qwen 7B	14/30	0.85 s	yes
9-frame contact sheet + Qwen2.5-VL	7/30	4.46 s	yes (19.24 MB)

Counts are answered honestly. The vault reports three numbers per label: `tracks` (raw ids, upper bound), `peak` (most same-label boxes in one frame, lower bound) and `physical` (after removing cross-label duplicates and one-frame transients, an estimate). On the cook clip, four "bowl" tracks resolve to **zero physical bowls** because two were plates double-labelled and one was a one-frame detection.

Key idea. The default answer path has no model in it. That makes it reproducible, cheap and fast on stage; the LLM is an optional phrasing layer, and the graph is the structured fallback.

3 Why it wins, and what to say

Claim	Proof
Cheap to keep	249 MB video → 95 KB memory; whole job dir still under 1.3 MB
Fast to query	0.05–0.06 ms vault p50; 0.89 ms cold on a 15-minute walk
No video at query	captions, events, counts and evidence answer the question; video is for playback only
Auditable	every answer carries a timestamped citation and an evidence still; frame chip seeks to it
Offline and robust	the deterministic path works with no LLM; the graph answers when captions miss
Coverage	caption windows cover the whole episode, including segments with no interaction row

Honest limits (say these before judges ask). Captions are lossy; fine motion, small text, audio and subtle social cues are lost. The VLM can hallucinate detail. Counts are estimates, not hand-labelled ground truth. Caption-first clips without a graph have no state/edge fallback. The 30-question suite is internal, not a public benchmark. The system is strongest on "what happened / where / when / how many / what is X holding", weakest on "why / exactly how / tiny object".

Judge Q&A cheat sheet

- **"Isn't this just captions?"** No: captions + a salience event ledger + graph counts + deterministic intents + evidence. Captions alone cannot answer "when", "how many" or "where" cleanly.
- **"Why not just use a video LLM?"** We do, at compile time. The win is query time: the baseline re-samples the video for every question; we compile once and answer in microseconds.
- **"Is 30/30 a benchmark?"** No — it is an internal regression suite covering location, temporal, count, negation and paraphrase cases. External benchmarks are next.
- **"What is lost?"** Fine motion, audio, tiny text, some rare objects. Evidence and citations make errors inspectable, not invisible.
- **"Does it need the video?"** No. That is the whole point; the source is kept only for playback and screenshots.

The 60-second demo

1. Show the 15-minute, 249 MB video and the 95 KB memory side by side.
2. Ask "where are my keys?". Show the answer, timestamp and frame chip.
3. Ask "what happened around 10 minutes?". Show the caption window and evidence still.
4. Ask a count question. Show `tracks / peak / physical` and the dedup example.
5. Stop the LLM and repeat a query: the deterministic answer still cites the graph.
6. Ask something the vault misses; show the labelled `graph-fallback` answer.

One line to close. "Every question about a long video normally means re-watching it. We compile the video into a 95 KB memory once, then answer in under a millisecond, with evidence — and the video is optional."

Not a replacement for a video LLM — the memory layer that makes one unnecessary for repeated questions.